

TrajectoryModel Architecture (Motion-FPN + Cross-Attention Kinematic V2)

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1 Overview

The current production model is the Motion-FPN + Cross-Attention Kinematic V2 planner, implemented in `model/CreateModel.py`. It predicts future ego kinematics from two consecutive front-camera frames and optional ego-history context. The model is built as a motion-aware image encoder plus a stepwise recurrent decoder that performs cross-attention over spatial memory at every prediction step.

2 Input and Output Specification

Item	Shape / Type	Notes
Current frame I_t	$(B, 3, 360, 640)$	RGB, normalized tensor
Previous frame I_{t-1}	$(B, 3, 360, 640)$	About 0.1 s before I_t
Ego history (optional)	$(B, 8, 3)$	[speed, accel, yaw_rate]
Prediction output	$(B, 12, 2)$	[speed _{m/s} , yawRate _{rad/s}]
Default horizon times	12 values	Non-uniform 3.0 s horizon; full list in Section 6

3 Architecture Blocks

3.1 Low-Level Utility Blocks

- **ConvGNAct**: convolution + GroupNorm + SiLU.
- **SEBlock**: squeeze-excitation channel recalibration.
- **Residual**: two ConvGNAct layers with residual skip, optional SE.
- **FiLM**: feature-wise affine conditioning from ego context.

3.2 MotionBackbone (Shared Weights)

Both frames are encoded by the same backbone.

- **Stem**: ConvGNAct(3 to base, stride 2) + Residual(base, SE).

- Stage1: ConvGNAAct(base to 2 base, stride 2) + Residual(2 base, SE).
- Stage2: ConvGNAAct(2 base to 3 base, stride 2) + Residual(3 base, SE), output at 1/8 scale.
- Stage3: ConvGNAAct(3 base to 4 base, stride 2) + Residual(4 base, SE), output at 1/16 scale.

3.3 MotionFpnEncoder

Per scale, the encoder builds motion-aware features from

$$[f_t, f_{t-1}, f_t - f_{t-1}]$$

then applies 1x1 fusion convolutions, top-down FPN fusion, and a final ConvGNAAct.

Key details:

- Two motion scales are fused: 1/8 and 1/16.
- Normalized coordinate channels (x, y) in $[-1, 1]$ are appended.
- FiLM uses ego-context to modulate the final fused map.
- Optional motion-map output is available for visualization.

Output tensor shape is $(B, \text{featDim} + 2, H/8, W/8)$. For 360x640 input, this is approximately $(B, \text{featDim} + 2, 45, 80)$.

3.4 EgoStateEncoder

A single-layer GRU encodes ego-history $(B, 8, 3)$ into a context vector $(B, 256)$. During training, ego context uses dropout (default `egoDropoutProb=0.15`) to reduce ego-only shortcut behavior.

3.5 CrossAttentionDeltaKinematicDecoder

This decoder replaced the old parallel transformer-vector head. It is autoregressive and predicts **delta kinematic updates** step by step.

Per prediction step:

- Spatial memory: encoder map is projected to hidden dimension, flattened, and layer-normalized.
- Time embedding: built from pair $(t_k, \Delta t_k)$.
- Query token uses recurrent hidden state, ego context, previous kinematic state, and time embedding.
- Cross-attention reads from spatial memory.
- A GRU updates recurrent state and a delta head outputs $(\Delta v, \Delta \dot{\psi})$.
- Current state is accumulated autoregressively.

Teacher forcing is one-step lagged to avoid current-step target leakage. Attention maps can be returned for analysis and visualization.

4 Temporal Rollout and Trajectory Integration

The model outputs speed and yaw-rate over non-uniform future offsets. Training and inference utilities integrate these kinematics as:

$$\psi_t = \psi_{t-1} + \dot{\psi}_t \Delta t_t$$

$$x_t = x_{t-1} + v_t \sin(\psi_t) \Delta t_t, \quad y_t = y_{t-1} + v_t \cos(\psi_t) \Delta t_t$$

where local coordinates are right-forward in meters.

5 Default Configuration (Current Family)

Parameter	Typical value
featDim	384
hiddenDim	768
baseChannels	48
numHeads	8
numLayers	4
predSteps	12
vectorTimes (s)	[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.85, 1.1, 1.35, 1.6, 2.3, 3.0]
model name	TrajectoryModel_MotionFpn_CrossAttentionKinematic_V2

6 Summary

The current model is a motion-conditioned, ego-aware, cross-attention recurrent planner. Compared to the legacy transformer-vector formulation, V2 shifts to kinematic-state rollout with per-step spatial re-attention, which improves temporal consistency and makes attention diagnostics more interpretable.